B3.2 Transport (Plant)

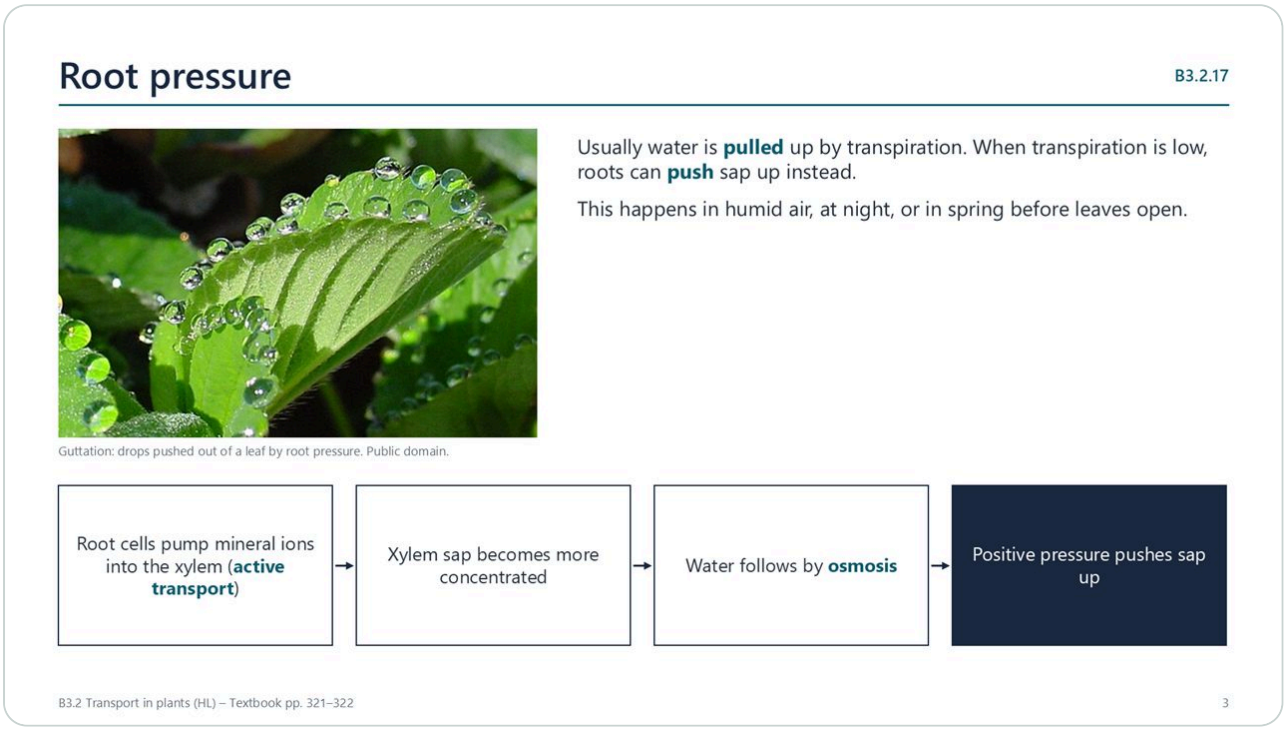
Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-b3-2-plant.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 321–325 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

ROOT PRESSURE AND PHLOEM

B3.2.17 Root pressure

pp. 321–322

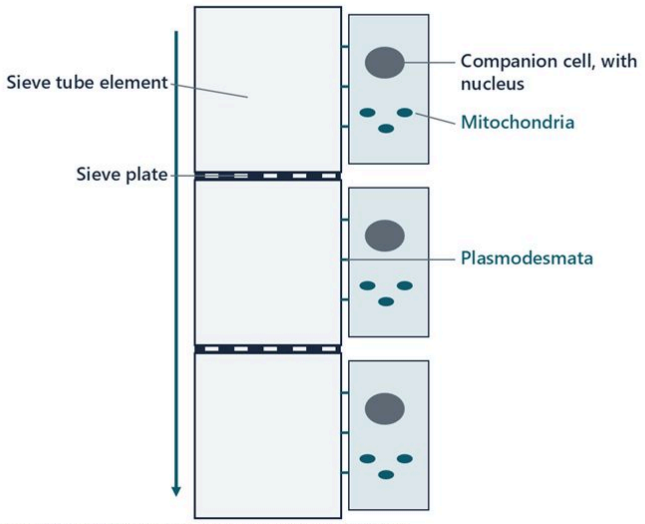


Root cells use ATP to pump mineral ions into the xylem by **active transport**. The xylem sap becomes more concentrated, so water follows by osmosis and a **positive pressure** builds that pushes sap upwards. Usually water is pulled up by transpiration, but root pressure matters when transpiration is low: in very humid air, at night, or in spring before deciduous leaves open. It can refill xylem vessels that have filled with air, and it causes **guttation**, drops of sap at the edges of leaves.

B3.2.18 Phloem and translocation

Sieve tubes and companion cells

B3.2.18



Translocation moves sap with sugars and amino acids from **sources** to **sinks**.

Sieve tube elements form the tube. Each has a **companion cell** that keeps it alive.

Drawn longitudinal section. The arrow shows sap flow, which can go either way.

B3.2 Transport in plants (HL) – Textbook pp. 322–325 5

Translocation moves sap with sugars and amino acids from **sources** (such as mature leaves) to **sinks** (such as roots, fruits and growing buds). **Sieve tube elements** have sieve plates with large pores, little cytoplasm, no nucleus and thick, rigid walls, so sap flows easily under pressure. **Companion cells** have a nucleus and many mitochondria, and are joined to the sieve tube by plasmodesmata, so they keep it alive and supply ATP for loading sucrose. Loading raises the pressure at the source and unloading lowers it at the sink, so sap flows from source to sink.

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